Data Science Capstone Project Report

GitHub: https://github.com/Nick-Charal/Applied-Data-Science-Capstone

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OUTLINE

- Executive Summary
- Introduction
- Methodology
- Results
  - Visualization – Charts
  - Dashboard
- Discussion
  - Findings & Implications
- Conclusion
- Appendix

EXECUTIVE SUMMARY

- Objective: To predict the success of SpaceX Falcon 9 first stage landings.
- Key Findings:
  - Success rates are improving over time, showing a correlation between success and number of flights.
  - Data analysis shows that launch sites are located near coasts.

INTRODUCTION

- SpaceX advertises Falcon 9 rocket launches with a cost of $62 million when other providers charge over $165 million.
- Key factor for this reduction of price is the reuse of the rocket's first stage.
- This project uses historical launch data to accurately predict if the first stage will land successfully.

METHODOLOGY

- Data Collection:
  - SpaceX API: Extracted useful information like Booster Version, Launch Sites, Payload data and landing outcomes.
  - Web Scraping: Collected historical launch records for Falcon 9 and Falcon Heavy Launches from Wikipedia.
- Data Wrangling:
  - Replaced missing values with average values and converted complex data into binary labels.
- Exploratory Data Analysis:
  - Performed database queries and trend visualization.
  - Utilized Folium to create interactive geographic analysis.
- Machine Learning:
  - Tested multiple models utilizing GridSearchCV for hyperparameter tuning.

RESULTS

- Trends:
  - Data analysis showed that there is a correlation between number of flights and success rate, regardless of payload mass.
- Geographic Analysis:
  - Launch sites are located near coasts to minimize costs for ocean landings.
- Model Performance:
  - All classification models reached a satisfactory level of accuracy over 70%.
  - KNN was the top performed with an accuracy of 90%.

DISCUSSION

- Built an interactive dashboard using Dash and Plotly.
- Utilized pie charts to visualize total successful launches for all sites or by site.
- Utilized scatter plots to visualize success rates vs payload mass by site

OVERALL FINDINGS & IMPLICATIONS

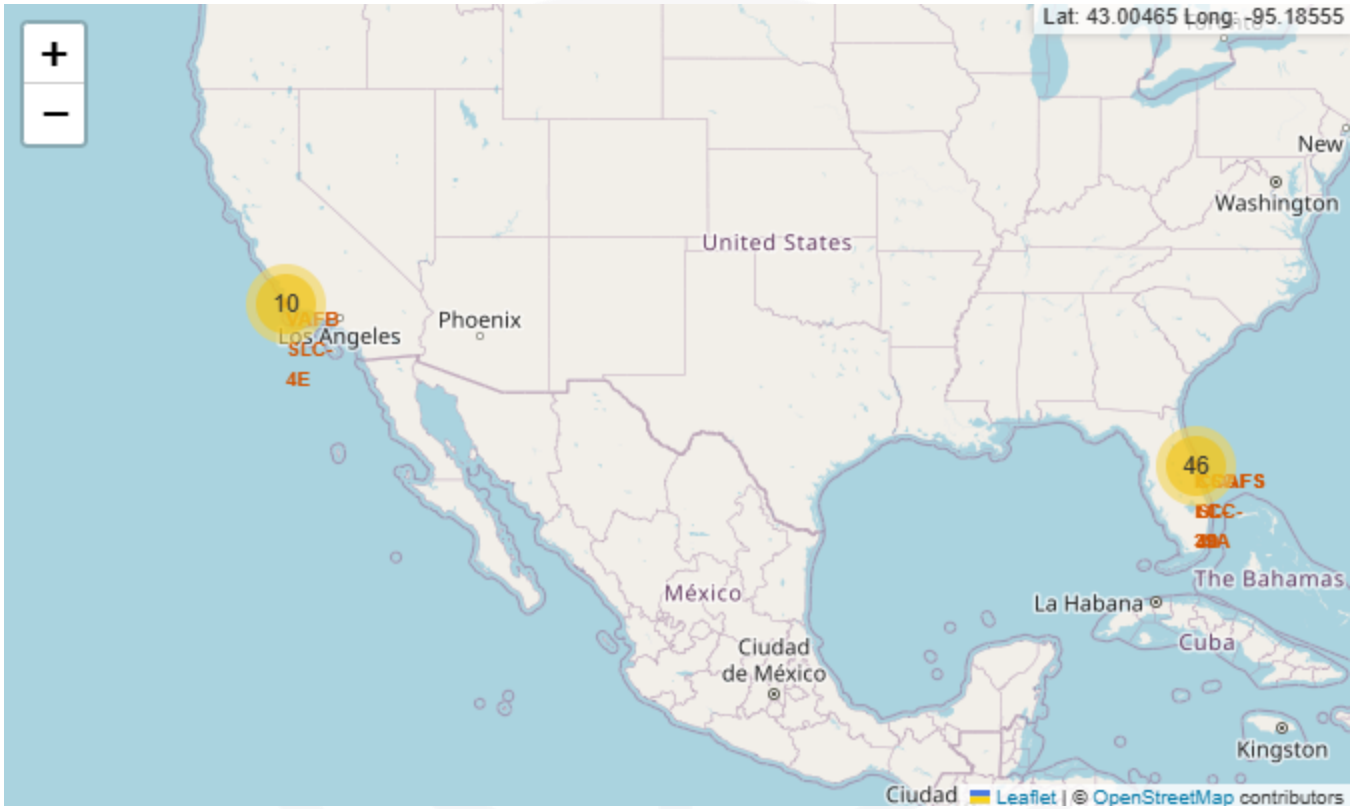
Findings

- Historical success rates have improved overall.

Implications

- Accurately predicting successful landings helps SpaceX to minimize their launch costs.

APPENDIX (Folium maps)



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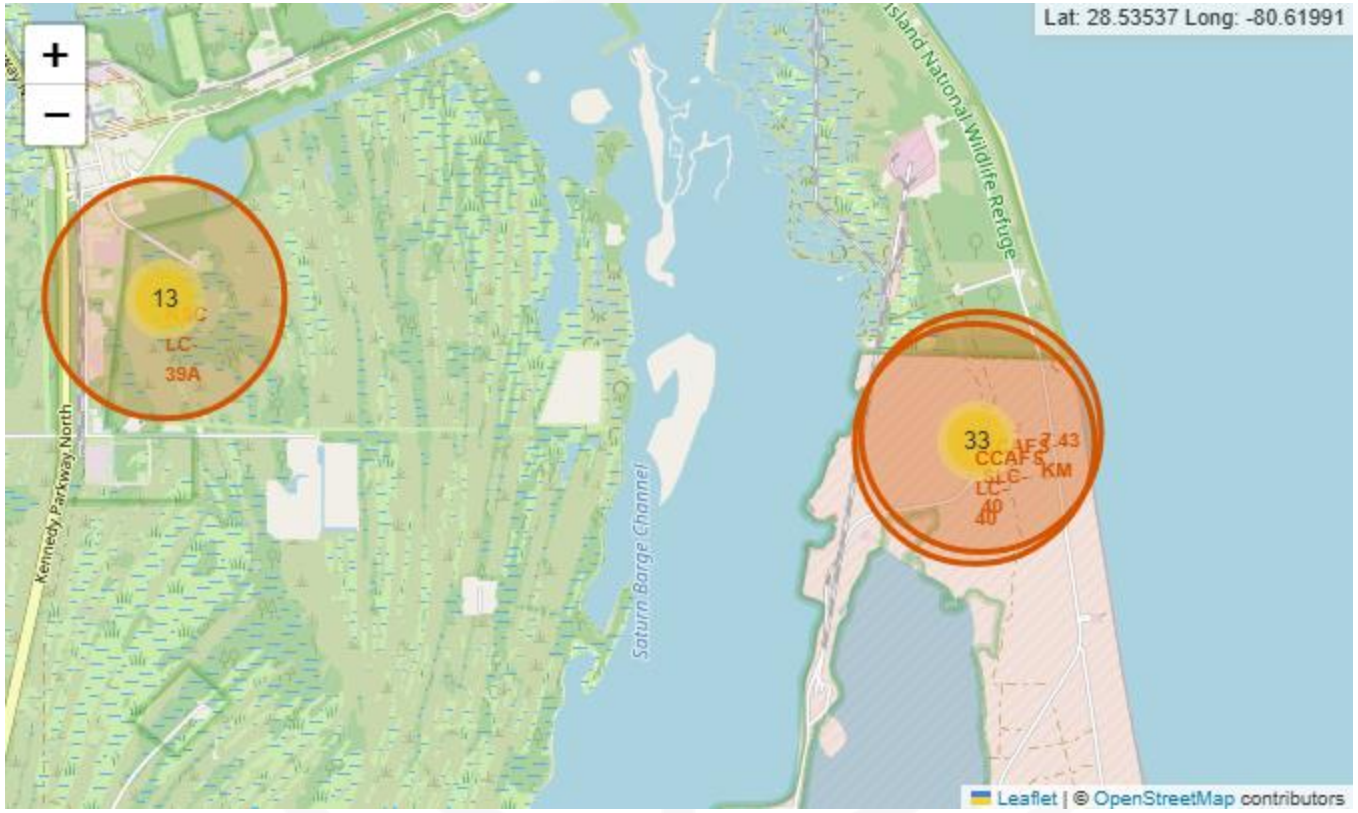

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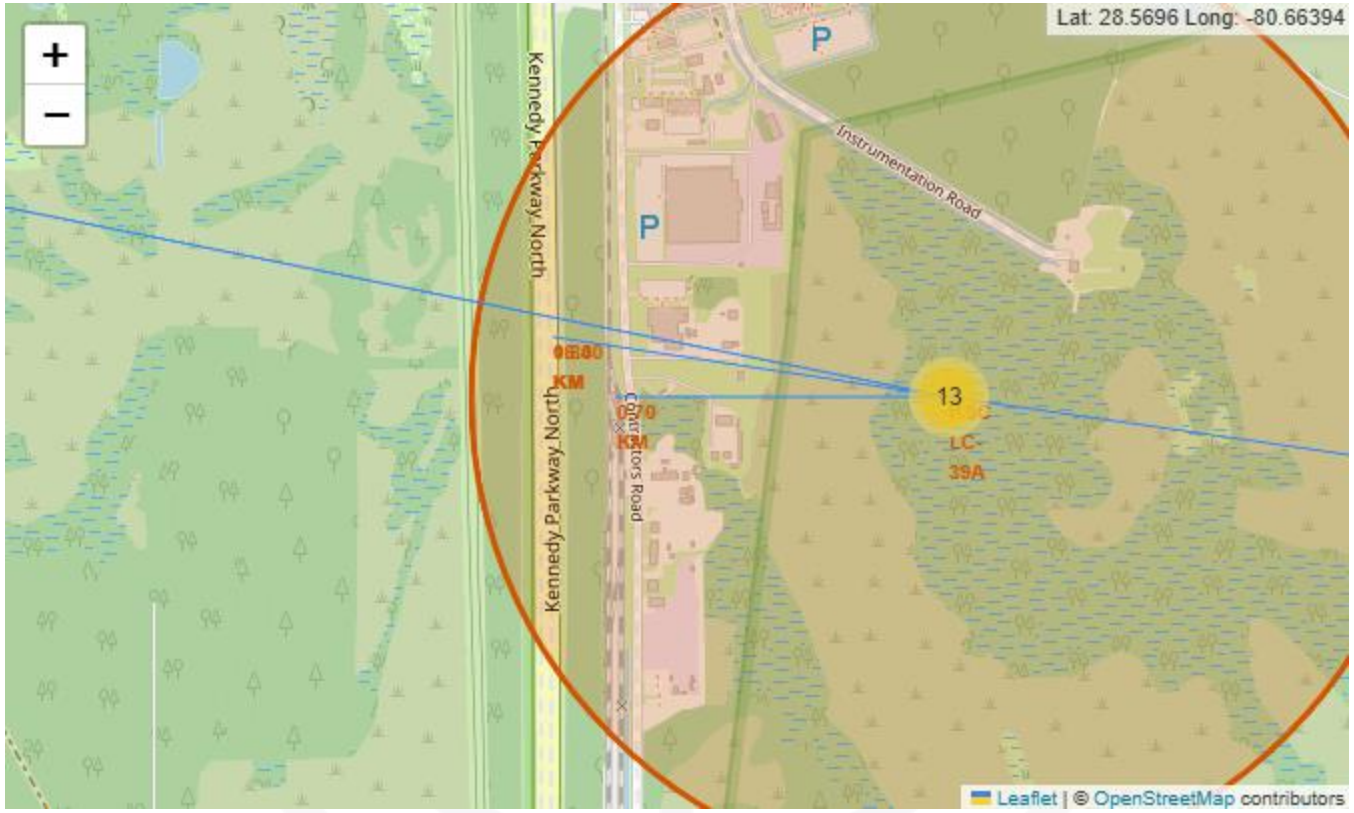
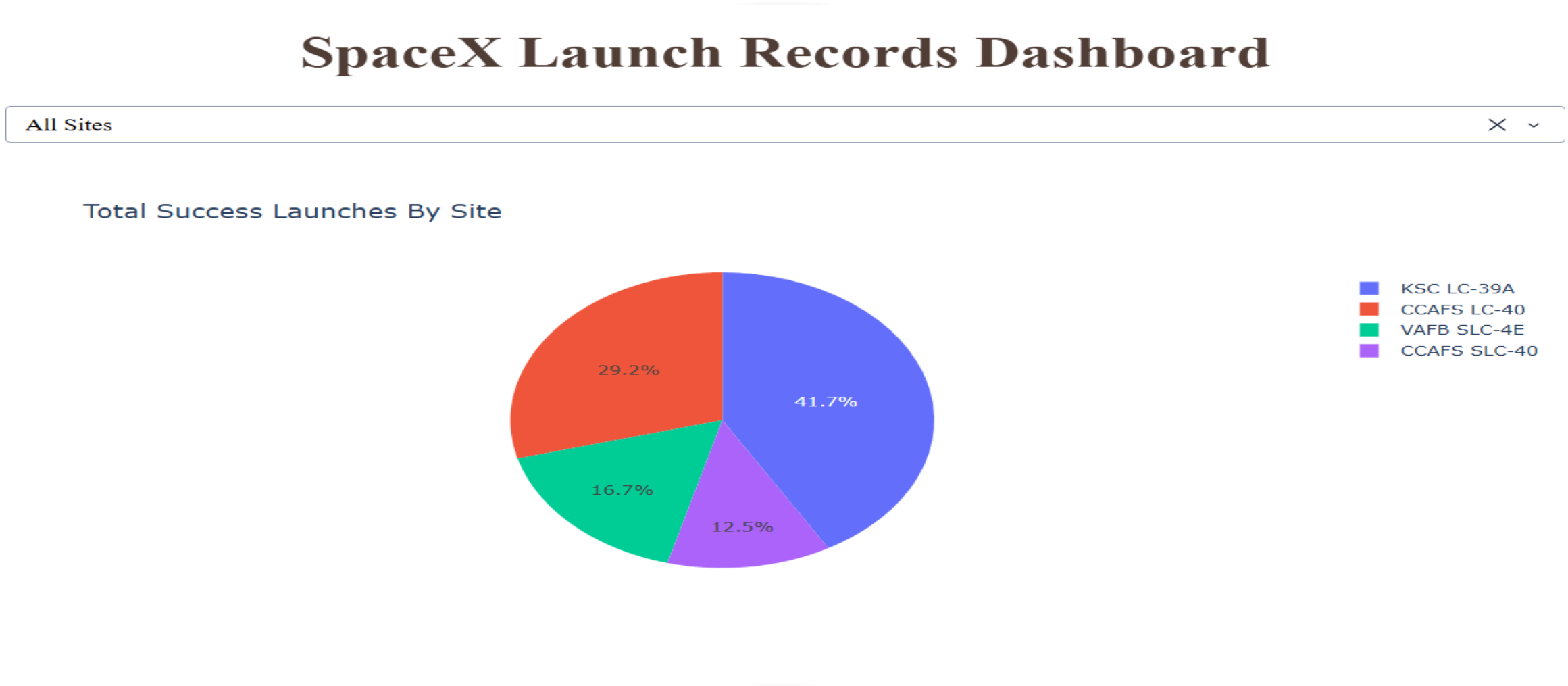


APPENDIX (Dashboard)

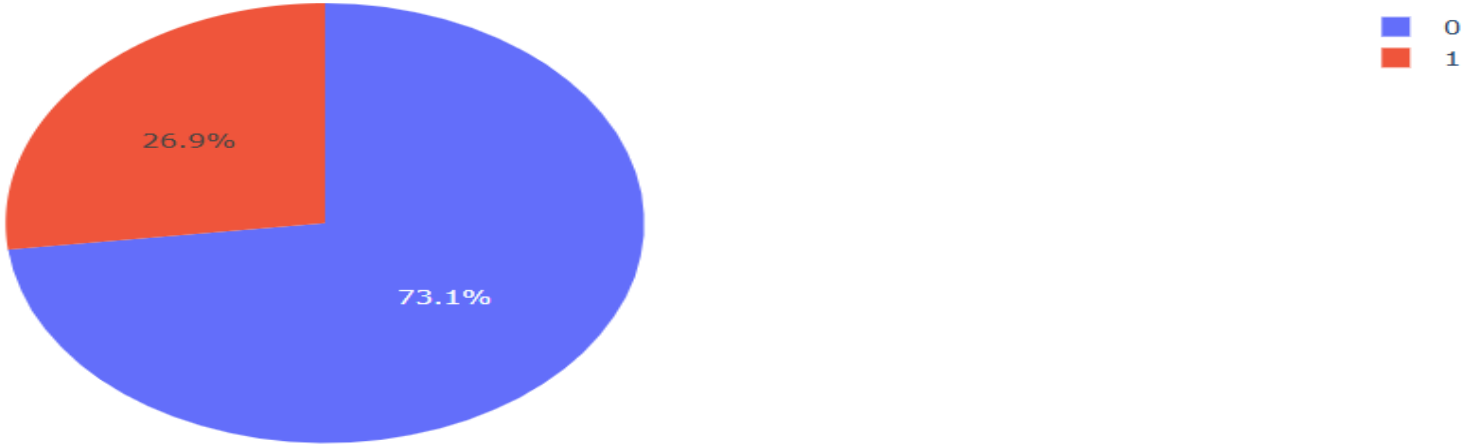


APPENDIX (Dashboard)

SpaceX Launch Records Dashboard

CCAFS LC-40

Total Success Launches for site CCAFS LC-40



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Payload range (Kg):

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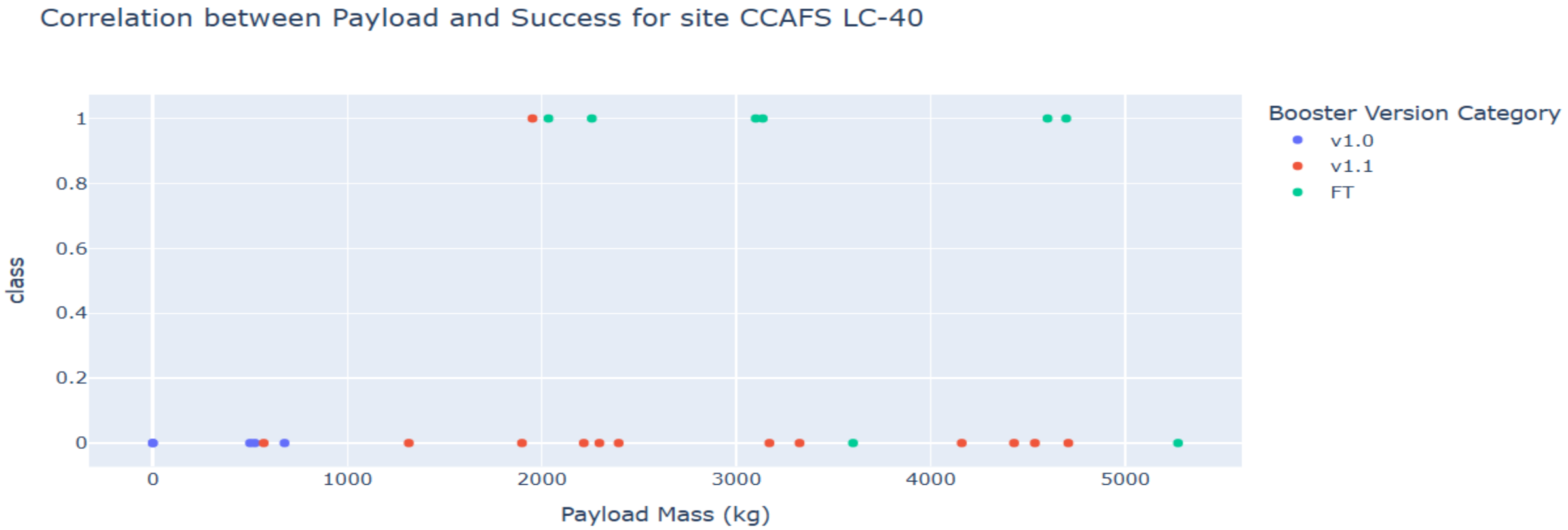
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